W4 *PRACTICE*

MANIPULATE BASIC WIDGETS

Learning objectives

- ✓ Start from an **empty Flutter project**
- ✓ **Hot Reload**
- ✓ Use Flutter **doctor**, **update**, create, run
- ✓ Use Flutter **Documentation**

- ✓ Manipulate **Scaffold**, Text, TextStyle, Radius
- ✓ Manipulate Colors palettes
- ✓ Manipulate **Container**, BoxDecoration, Center, EdgeInsets, Column

No AI tools allowed to solve this practice

How to submit?

- ✓ **Push** your final code on **your GitHub repository**
- ✓ Then **attach the GitHub path** to the MS Team assignment and **turn it in**

Are you lost?

Read the following documentation to be ready for this practice:

https://api.flutter.dev/flutter/material/Scaffold-class.html
https://docs.flutter.dev/ui/widgets/text
https://api.flutter.dev/flutter/widgets/Center-class.html
https://api.flutter.dev/flutter/painting/EdgeInsets-class.html
https://api.flutter.dev/flutter/widgets/Column-class.html
https://api.flutter.dev/flutter/widgets/Container-class.html
https://api.flutter.dev/flutter/painting/BoxDecoration-class.html

EX 1 – Scaffold, Text

In this exercise, you will be working with the following widgets

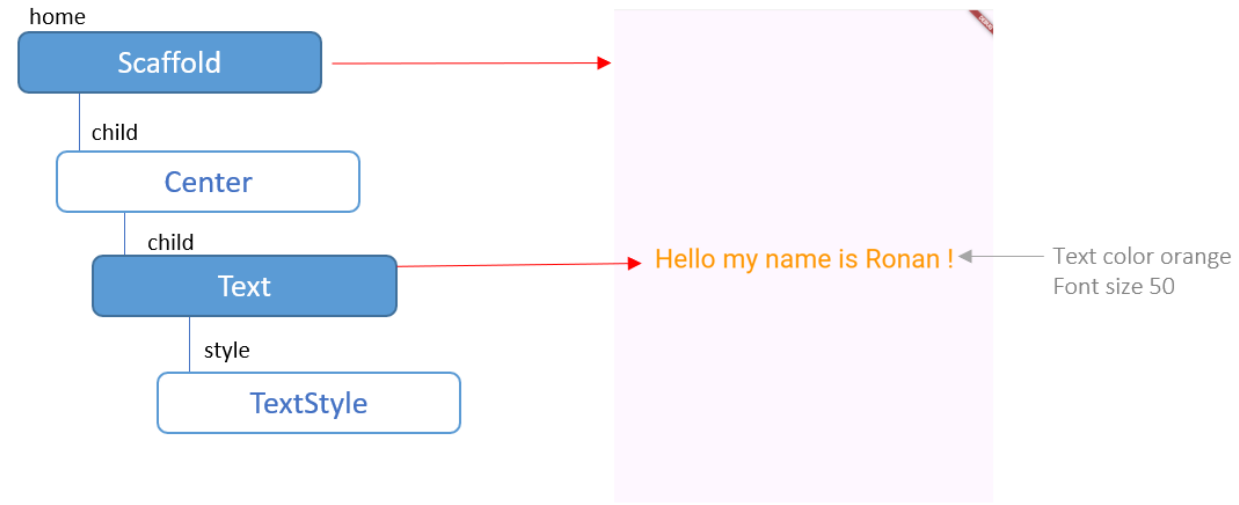
- **Scaffold:** Provides a basic structure for UI, like app bar, drawer, etc.
- **Center**: Aligns a child widget to the center
- **Text**: Displays text.

And the following classes

- **TextStyle**: Styles text appearance.

You need to produce the following mockup and widget structure:

You are free to customize it!



To start

Open `W3-S2/EX-1/main.dart`

```
void main() {
  runApp(
    const MaterialApp( ),
  );
}
```

EX 2 – Container, Insets, BoxDecoration

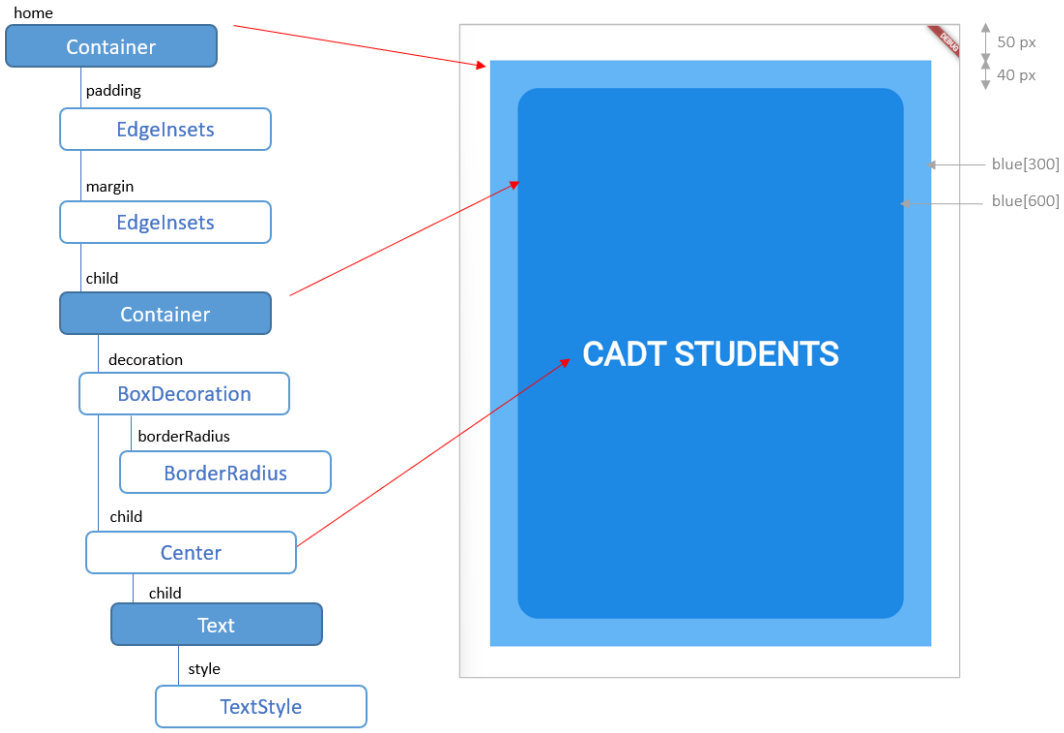
In this exercise, you will be working with the following widgets

- **Container**: A customizable box for layout and styling.
- **Text**: Displays text.
- **Center**: Aligns a child widget to the center

And the following classes

- **EdgeInsets**: Sets padding or margin.
- **BoxDecoration**: Styles a container's background, border, etc.
- **BorderRadius**: Rounds container corners.
- **TextStyle**: Styles text appearance.

You need to produce the following mockup and widget structure:

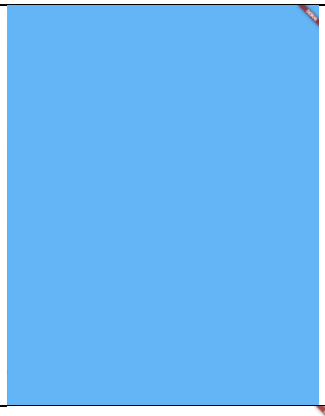


We recommend you to follow **the bellow steps** to be able to understand each widget in details. However, you are free to choose your favorite workflow!

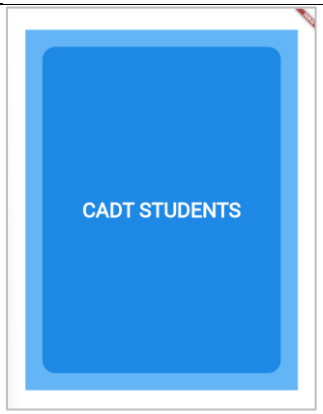
<u>To start</u>

Open `W3-S2/EX-2/main.dart`

```
void main() {
  runApp(MaterialApp(
    home: Container( ),
  ));
}
```

STEP 1		Just a blue container for the home…
		
		`MaterialApp`
		`  home: Container`

STEP 2		Add an inner container and some padding margins and box decoration
		
		`MaterialApp`
		`  home: Container`
		`  margin: EdgeInsets`
		`  padding: EdgeInsets`
		`  child: Container`
		`  decoration: BoxDecoration`

STEP 3		Add the Text, centered, and style it.
		
		`MaterialApp`
		`  home: Container`
		`  margin: EdgeInsets`
		`  padding: EdgeInsets`
		`  child: Container`
		`  decoration: BoxDecoration`
		`  child: Center`
		`  child: Text`
		`  style TextStyle`

In this exercise, we haven't used the Scaffold widget: **Why**? What is the **purpose of this widget**?

EX 3 – Column Gradient

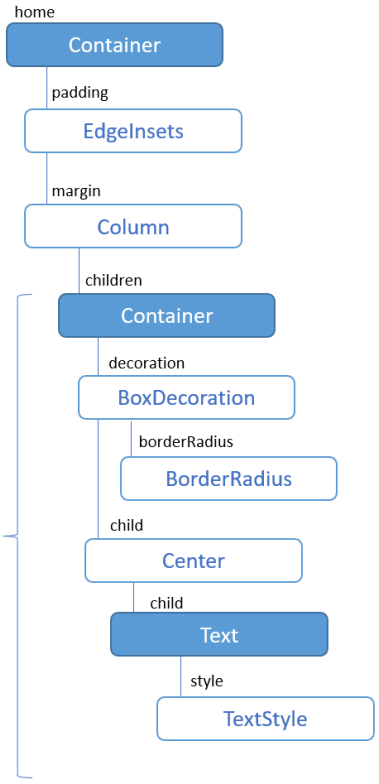
In this exercise, you will be working with the following widgets

- **Container**: A customizable box for layout and styling.
- **Column** Arranges widgets vertically in a column.

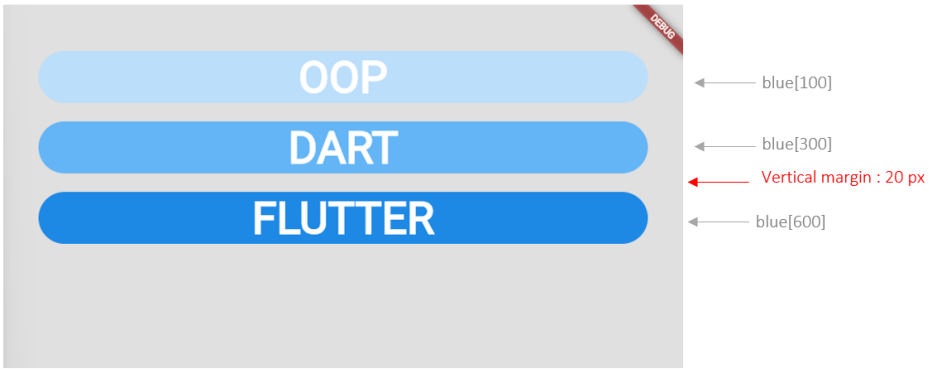
And he following classes

- **EdgeInsets**: Sets padding or margin.
- **BoxDecoration**: Styles a container's background, border, etc.

You need to produce the following mockup and widget structure:

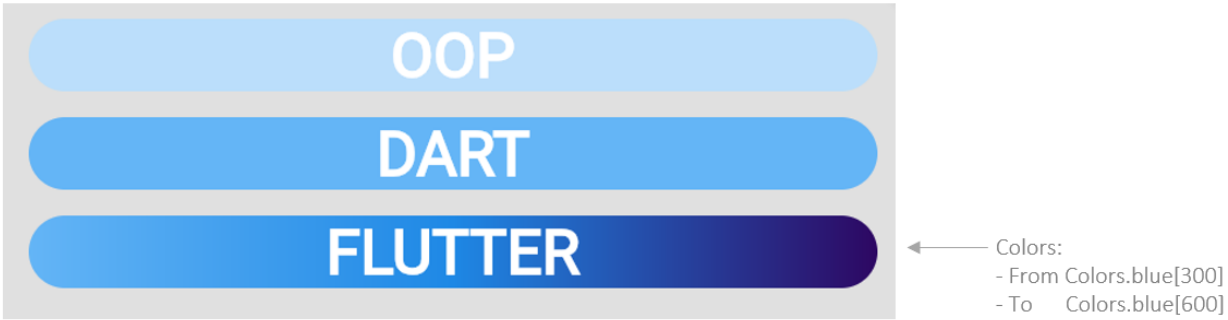


X3



BONUS

In the last item, add a [linear gradient] as specified bellow



EX 4 – From mockup to code (PREPARE FOR TEST)

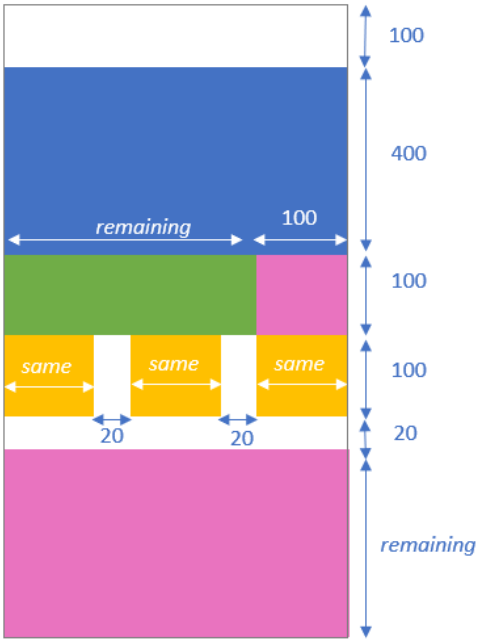
Build the bellow UI with the bellow screen:

Widgets:

- Column
- Row
- Expanded
- Container
- SizedBox

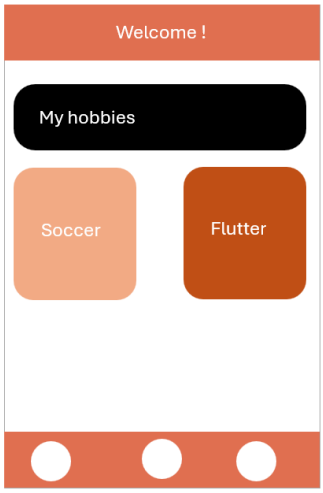
Configuration classes

- EdgeInsets
- CrossAxisAlignment
- MainAxisAlignment



EX 5 – Your welcome page ***(PREPARE FOR TEST)***

Your goal is to combine all the widgets you have learned and create a **welcome page of your choice,**



In this exercise, you can use the following widgets

1. Basic Structure
   - **Scaffold**: Provides a basic structure for a mobile screen

2. Layout and Alignment
   - **Row**: Arranges widgets horizontally
   - **Column**: Arranges widgets vertically
   - **Center**: Aligns a child widget to the center
   - **Expanded**: Makes a child of a Row, Column, or Flex fill the available space
   - **SingleChildScrollView**: Makes a single child scrollable

3. Spacing and Styling
   - **Container**: A customizable box for layout and styling
   - **SizedBox**: Defines a fixed width or height or adds space between widgets
   - **Padding**: Adds empty space around a widget's child

4. Content
   - **Text**: Displays text.

And the following classes

- TextStyle: Styles text appearance.
- BoxDecoration Styles a container's background, border, etc.
- EdgeInsets : Sets padding or margin.

💡 You want to draw a circle? You can customize the contain box decoration with a circle shape:

```
Container(
  width: 100.0,
  height: 100.0,
  decoration: BoxDecoration(
    shape: BoxShape.circle,
  ),
)
```